

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	PRITHIYANKA SHREE M	Reg. No.:	192521144
Section / Batch:	IT-2025	Date:	28-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

A robotics system processes coordinate points to determine object boundaries using brute force convex hull computation. a) Explain how convex hull is determined. b) Analyze the time complexity. c) Discuss scalability concerns.

Code of Conduct

I M PRITHIYANKA SHREE M (192521144) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M PRITHIYANKA SHREE

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex realworld problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & DecisionMaking Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

QUESTION:

A robotics system processes coordinate points to determine object boundaries using brute-force convex hull computation. a) Explain how convex hull is determined. b) Analyze the time complexity. c) Discuss scalability concerns.

CODE (IMPLEMENTATION):

```
import matplotlib.pyplot as plt

# Function to find orientation
def orientation(p, q, r):
    val = (q[1] - p[1]) * (r[0] - q[0]) - (q[0] - p[0]) * (r[1] - q[1])
    if val == 0:
        return 0      # Collinear
    elif val > 0:
        return 1      # Clockwise
    else:
        return 2      # Counterclockwise

# Brute Force Convex Hull
def brute_force_convex_hull(points):
    n = len(points)
    hull = []

    if n < 3:
        return []

    for i in range(n):
        for j in range(i + 1, n):
            left = 0
            right = 0

            for k in range(n):
                if k == i or k == j:
                    continue

                o = orientation(points[i], points[j], points[k])

                if o == 2:
                    left += 1
                elif o == 1:
                    right += 1

            if left == 0 or right == 0:
                if points[i] not in hull:
                    hull.append(points[i])
                if points[j] not in hull:
                    hull.append(points[j])
```

```

# ----- Main Program -----

n = int(input("Enter the number of points: "))

points = []

for i in range(n):
    x = int(input(f"Enter x{i+1}: "))
    y = int(input(f"Enter y{i+1}: "))
    points.append((x, y))

hull = brute_force_convex_hull(points)

print("\nConvex Hull Points:")
for p in hull:
    print(p)

# Plotting
x = [p[0] for p in points]
y = [p[1] for p in points]

plt.scatter(x, y, color="blue")

for p in points:
    plt.text(p[0], p[1], str(p))

if len(hull) > 1:
    hx = [p[0] for p in hull] + [hull[0][0]]
    hy = [p[1] for p in hull] + [hull[0][1]]
    plt.plot(hx, hy, color="red")

plt.title("Brute Force Convex Hull")
plt.xlabel("X")
plt.ylabel("Y")
plt.grid(True)
plt.show()

# Sort hull points
cx = sum(x for x, y in hull) / len(hull)
cy = sum(y for x, y in hull) / len(hull)

import math
hull.sort(key=lambda p: math.atan2(p[1] - cy, p[0] - cx))

return hull

```

INPUT:

```

Enter the number of points: 7
Enter x1: 0
Enter y1: 3
Enter x2: 2
Enter y2: 2
Enter x3: 1
Enter y3: 1
Enter x4: 2
Enter y4: 1
Enter x5: 3
Enter y5: 0
Enter x6: 0
Enter y6: 0
Enter x7: 3
Enter y7: 3

```

OUTPUT/RESULT:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
```

```
= RESTART: C:/Users/Hello/AppData/Local/Programs/Python/Python314/assesment 2.py
```

```
Enter the number of points: 7
```

```
Enter x1: 0
```

```
Enter y1: 3
```

```
Enter x2: 2
```

```
Enter y2: 2
```

```
Enter x3: 1
```

```
Enter y3: 1
```

```
Enter x4: 2
```

```
Enter y4: 1
```

```
Enter x5: 3
```

```
Enter y5: 0
```

```
Enter x6: 0
```

```
Enter y6: 0
```

```
Enter x7: 3
```

```
Enter y7: 3
```

```
Convex Hull Points:
```

```
(0, 0)
```

```
(3, 0)
```

```
(3, 3)
```

```
(0, 3)
```

